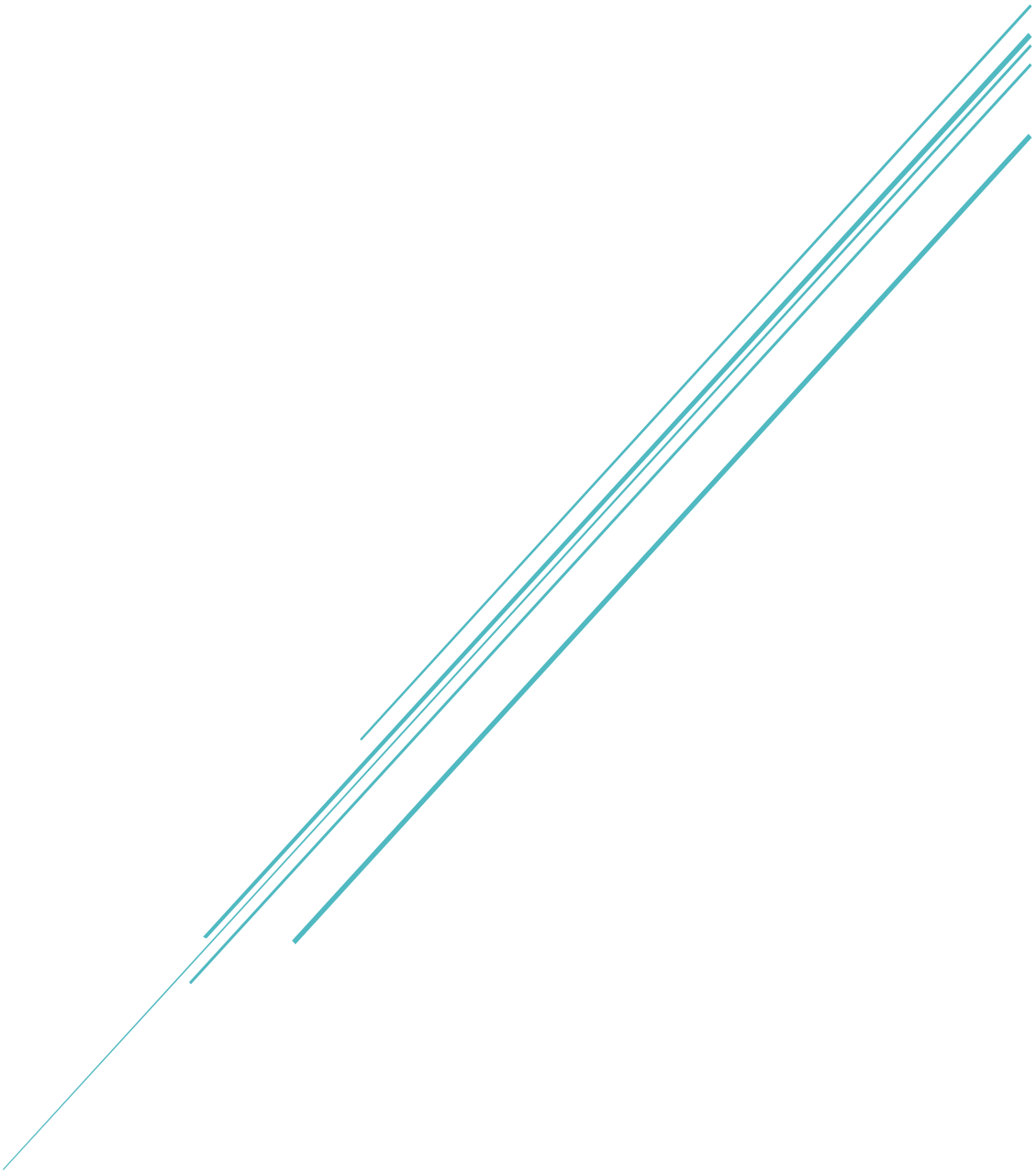
ASSESSMENT-2

Network Security, Maintenance & Troubleshooting Procedures



CompTia N+

Section A — Concept Application

S-1 SCENARIO : Nexus Innovations is purchasing new computers for employees who use programming software, web browsers, and video conferencing applications at the same time.

Question: What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?

To ensure good computer performance and hardware compatibility, the following factors should be considered:

1. CPU (Processor):

- Choose a multi-core processor for multitasking.
- Select a processor with good speed.
- Intel Core i5/i7 or AMD Ryzen 5/7 are good options.

2. RAM (Memory):

- Choose at least 16 GB RAM.
- 32 GB RAM is better for heavy multitasking.
- Check RAM compatibility with the motherboard.

3. Storage:

- Choose an SSD for faster performance.
- 512 GB or 1 TB storage is recommended.
- NVMe SSD provides faster data access.

4. Hardware Compatibility:

- Ensure the CPU is supported by the motherboard.

- Check RAM type and speed compatibility.
- Make sure the motherboard supports the selected SSD.

Conclusion:

A good CPU, sufficient RAM, and fast SSD help employees run programming software, web browsers, and video conferencing applications smoothly at the same time.

S-2 SCENARIO : After installing antivirus software on all startup office computers, employees notice that their systems have become slower.

Question: Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

Antivirus software protects computers from viruses and malware, but it also uses system resources, which can slow down the computer.

1. CPU (Processor):

- Antivirus continuously scans files and programs.
- It uses CPU processing power.
- Other applications may run slowly.

2. RAM (Memory):

- Antivirus uses RAM to run in the background.
- Less RAM is available for other applications.
- Running multiple programs may become slower.

3. Storage (HDD/SSD):

- Antivirus scans files stored on the drive.
- Frequent file reading and writing increases disk activity.
- Opening files and applications may take more time.

Conclusion:

Antivirus software improves computer security, but high CPU, RAM, and storage usage can reduce system performance and make the computer slow.

S-3 SCENARIO: A USB keyboard and printer work properly on one computer but do not work correctly on another computer in the startup office.

Question: Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

To troubleshoot USB keyboard and printer problems caused by driver incompatibility or operating system issues, follow these steps:

1. Check USB Connection:

- Check that the USB cables are properly connected.
- Try different USB ports.

2. Check Device Manager:

- Open Device Manager.

- Check for yellow warning signs near the keyboard or printer.

3. Update Drivers:

- Download the latest compatible drivers from the manufacturer's official website.
- Install the drivers.

4. Reinstall Drivers:

- Uninstall the faulty drivers.
- Install them again.

5. Restart Computer:

- Restart the computer after installing the drivers.

6. Check Operating System:

- Install pending Windows updates.
- Run Windows Troubleshooter.
- Use the sfc /scannow command to check and repair corrupted system files.

Conclusion:

These troubleshooting steps help fix driver and operating system issues so that the USB keyboard and printer work properly.

S-4 SCENARIO : Nexus Innovations experiences frequent power cuts and voltage fluctuations.

Question: Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations.

A UPS and Surge Protector help protect computer hardware and prevent data loss during power cuts and voltage fluctuations.

1. UPS (Uninterruptible Power Supply):

- Provides battery backup during power cuts.
- Keeps the computer running for a short time.
- Allows users to save their work.
- Helps prevent data loss due to sudden shutdowns.



2. Surge Protector:

- Protects computers from sudden voltage increases.
- Prevents damage to hardware like the motherboard and power supply.
- Reduces the risk of electrical damage.



Conclusion:

Using both a UPS and Surge Protector helps protect computer hardware, maintain power supply, and reduce the risk of data loss during power problems.

S-5 SCENARIO : New employees at Nexus Innovations report that they can reach internal systems (e.g. the application server) by IP address but not by name, and their devices sometimes fail to obtain a valid IP address when they connect to the network.

Question: Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described.

DNS and DHCP work together to help devices communicate properly on a network.

1. DNS (Domain Name System):

- Converts domain names into IP addresses.
- Helps users access servers using names instead of IP addresses.
- Example: appserver.local → 192.168.1.10.

2. DHCP (Dynamic Host Configuration Protocol):

- Automatically assigns IP addresses to network devices.
- Provides Subnet Mask, Default Gateway, and DNS Server Address.
- Helps devices connect to the network easily.

3. How DNS and DHCP Work Together:

- DHCP assigns an IP address and DNS server address to the employee's computer.
- When the employee enters a server name, DNS converts it into an IP address.
- The computer uses that IP address to connect to the server.

4. Most Likely Misconfigured Services:

- DNS Issue: The server opens by IP address but not by name, indicating a possible DNS configuration problem.
- DHCP Issue: Devices sometimes fail to obtain a valid IP address, indicating a possible DHCP configuration problem.

Conclusion:

Both DNS and DHCP may be misconfigured. DNS helps resolve names into IP addresses, while DHCP automatically provides IP configuration to devices.

S-6 SCENARIO : After the IT team installs a new firewall at Nexus Innovations, employees can access the internet but are unable to reach the critical internal application server.

Question: Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

To troubleshoot the firewall issue, follow these steps:

1. Check Network Connectivity:

- Ping the internal application server using its IP address.
- Check whether the server is reachable.

2. Check Firewall Rules:

- Open the firewall configuration.
- Check whether any rule is blocking traffic to the internal server.

3. Check Firewall Logs:

- Review firewall logs to identify blocked or denied traffic.
- Find the source and destination IP addresses.

4. Check IP Address and Port:

- Verify the application server's IP address.
- Check the required port, such as HTTP (80) or HTTPS (443).

5. Modify Firewall Rules:

- Create an appropriate Allow Rule for authorized traffic.
- Allow only the required IP addresses and ports.

6. Test the Connection:

- Try accessing the application server again.
- Confirm that the application is working properly.

Conclusion:

By checking firewall rules and logs, the IT team can identify blocked traffic and create the correct Allow Rules to restore access to the internal application server without reducing network security.

Section B — Practical Application

Task 1: Workstation Assembly

Physically assemble a workstation for the startup office and document the component installation sequence.

- Install CPU, RAM, storage, and other core components into the chassis in the correct sequence.
- Connect all required internal cables (power, data, front panel).
- Document each installation step in order. System must pass POST (Power-On Self-Test) after assembly.

The following steps are used to assemble a workstation for the startup office:

1. Install CPU:



- Place the CPU carefully into the motherboard socket.
- Lock the CPU properly.
- Apply thermal paste and install the CPU cooler if required.

2. Install RAM:



1.Open the computer case, check the memory slot position, discharge static electricity from your hands, and clean the dust in the slot



2.Open the latches on both sides of the memory slot



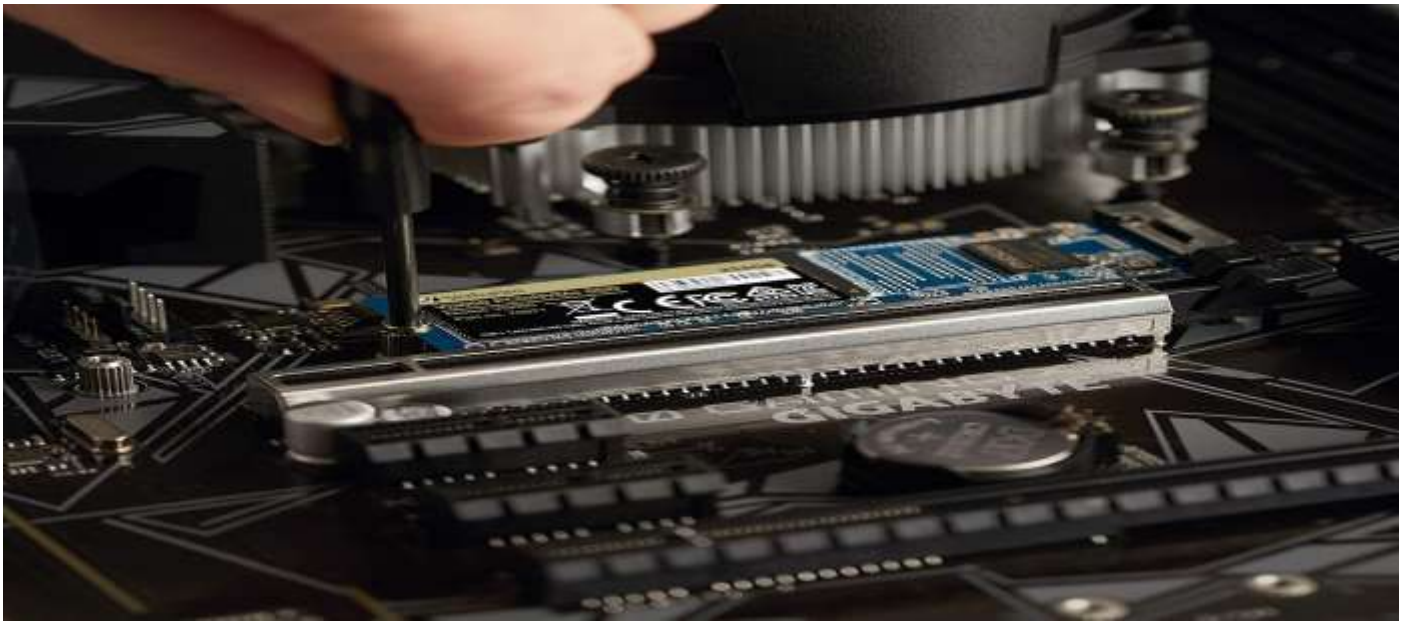
3.Align with the memory key notch, if it doesn't fit, rotate the memory module



4.After aligning with the key notch, press down on the memory module until a "click" sound is heard, and che c if the golden fingers are exposed. If not, the installation is successful

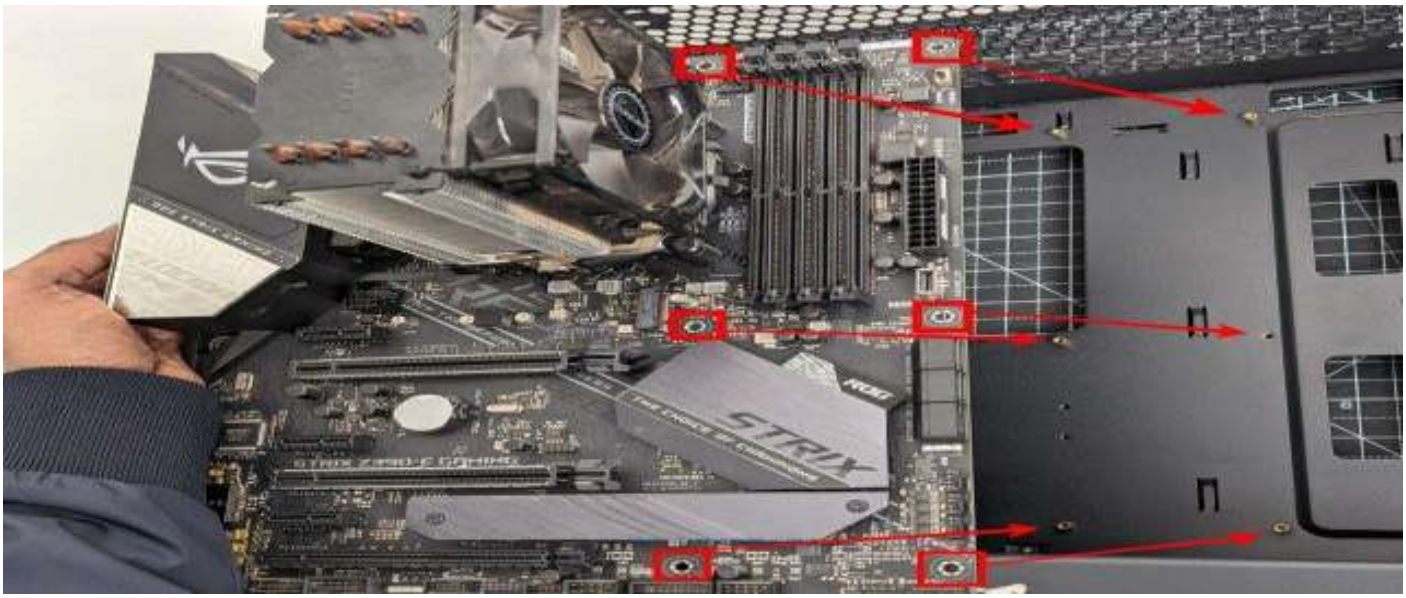
- Insert the RAM into the correct motherboard slots.
- Press gently until the clips lock.

3. Install Storage:



- Install the SSD or HDD in the correct slot.
- Connect SATA data and power cables if required.

4. Install Motherboard:



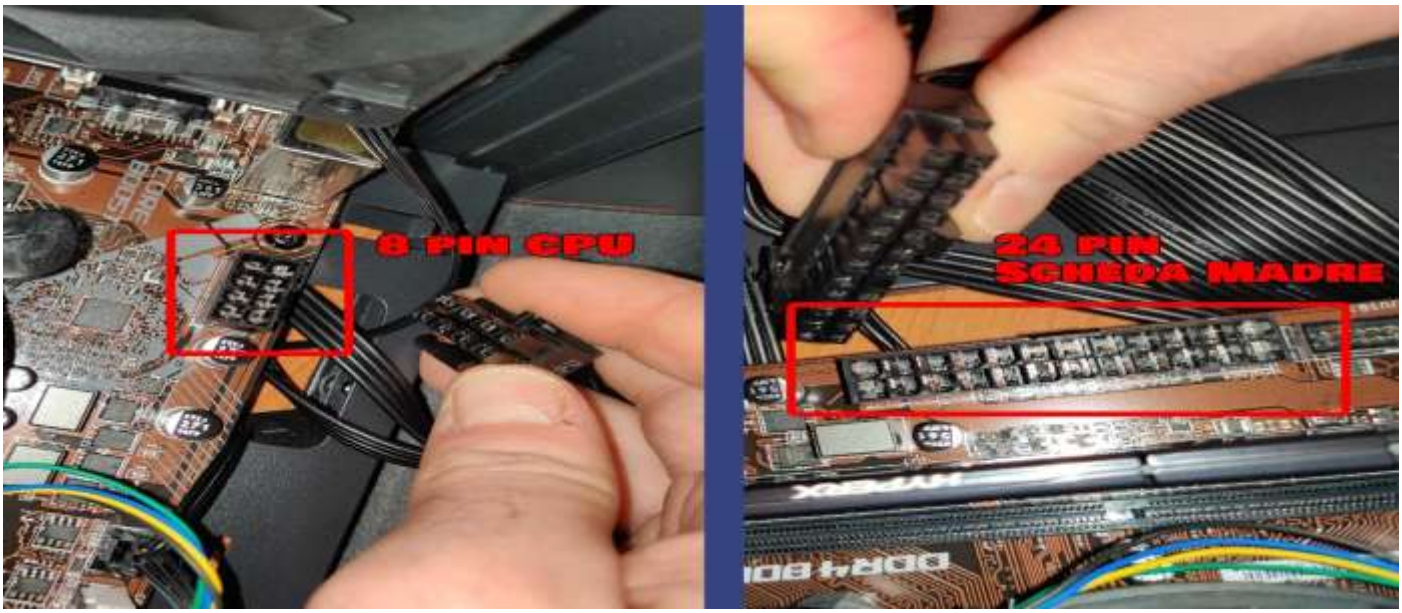
- Place the motherboard inside the cabinet.
- Secure it with screws.

5. Install SMPS (Power Supply):



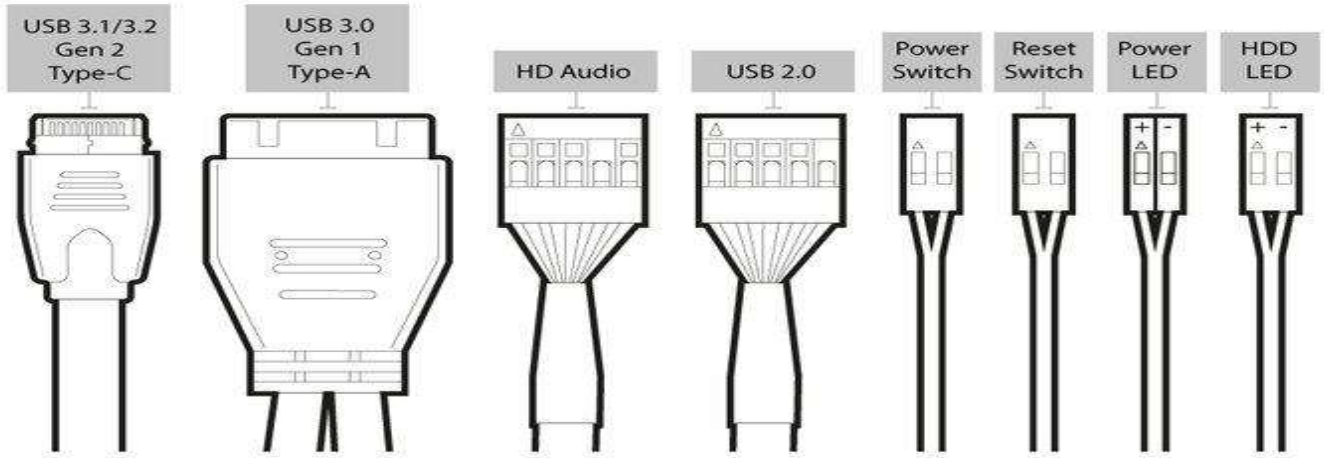
- Fit the SMPS inside the cabinet.
- Secure it with screws.

6. Connect Internal Cables:



- Connect the 24-pin motherboard power cable.
- Connect the CPU power cable.
- Connect storage power and data cables if required.

7. Connect Front Panel Cables:



- Connect Power Switch and Reset Switch.
- Connect Power LED and HDD LED.
- Connect front USB and Audio cables.

8. Final Checking:



- Check all components and cable connections.
- Ensure that no cable is loose.

9. Perform POST Test:



- Connect the monitor, keyboard, and power cable.
- Turn on the computer.
- Check whether the system starts without hardware errors.
- Confirm that the POST test is successful.

Safety Precautions:

- Disconnect the power supply before assembly.
- Use an anti-static wrist strap.
- Handle all components carefully.

Conclusion:

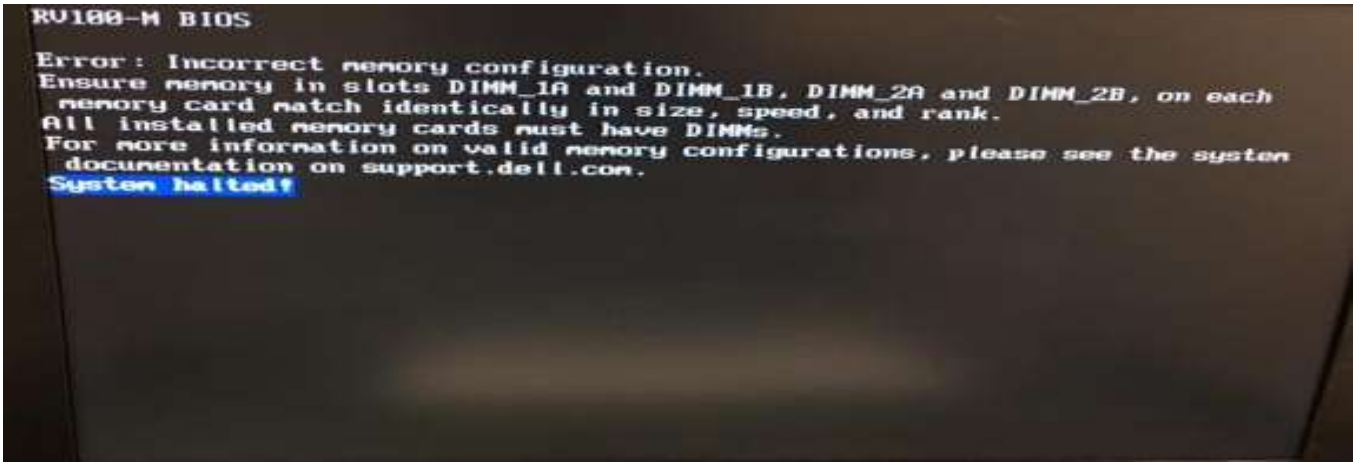
A workstation is successfully assembled when all components are properly installed, all cables are connected correctly, and the system passes the POST test.

Task 2: RAM Mismatch Troubleshooting

Troubleshoot and resolve a RAM mismatch error within the BIOS/UEFI settings on a startup office workstation.

- Identify the RAM mismatch error via BIOS/UEFI POST messages.
- Determine the root cause (mismatched speed, timing, or capacity).
- Resolve the mismatch by reconfiguring or reseating the RAM correctly.
- Confirm the system boots normally after the fix.

1. Identify RAM Error:



- Restart the computer and enter BIOS/UEFI.
- Check POST messages for RAM or memory errors.

2. Find the Root Cause:



- Check RAM speed, timing, and capacity.
- Identify any compatibility issues between RAM modules.

3. Reseat the RAM:



1.Open the computer case, check the memory slot position, discharge static electricity from your hands, and clean the dust in the slot



2.Open the latches on both sides of the memory slot



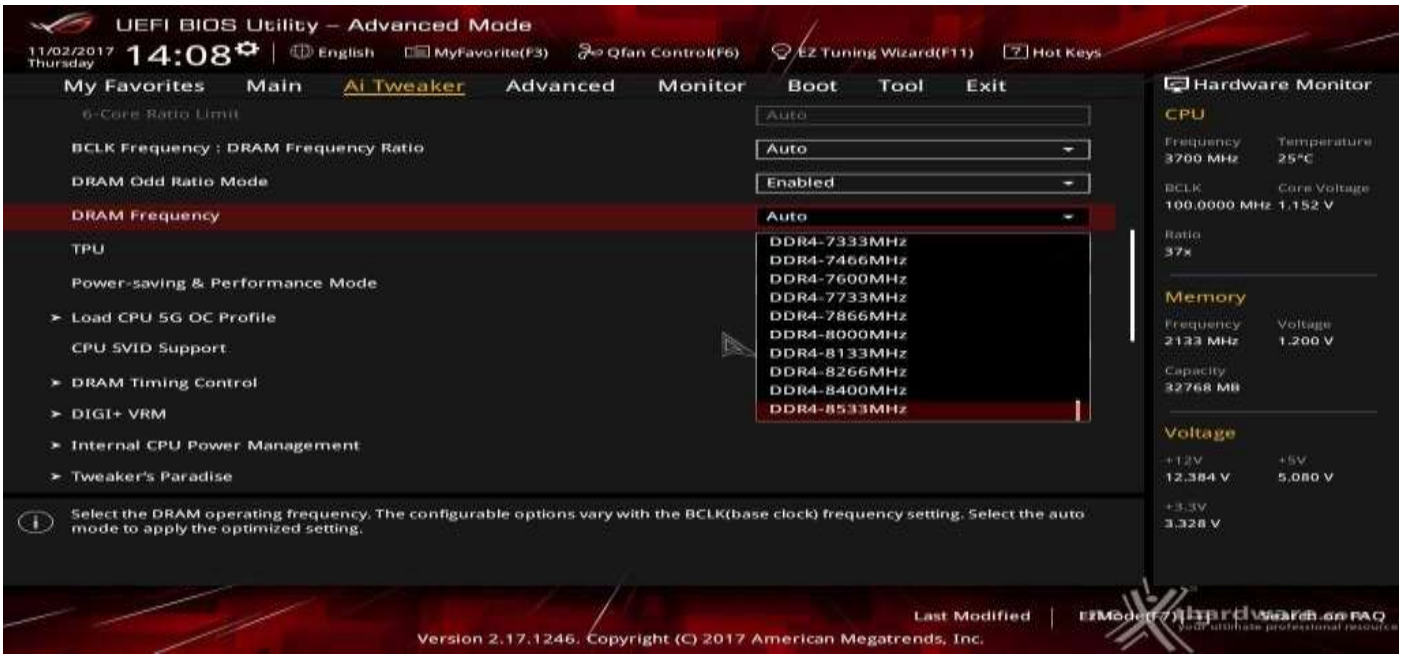
3.Align with the memory key notch, if it doesn't fit, rotate the memory module



4.After aligning with the key notch, press down on the memory module until a "click" sound is heard, and che c if the golden fingers are exposed. If not, the installation is successful

- Turn off the computer and disconnect the power cable.
- Remove the RAM carefully.
- Reinsert it properly into the correct motherboard slots.

4. Reconfigure RAM Settings:



- Open BIOS/UEFI settings.
- Set compatible RAM speed and timing.
- Disable XMP/EXPO if necessary.
- Save the settings.

5. Test the System:

- Restart the computer.
- Check whether all RAM is detected correctly.
- Confirm that the system boots normally without errors.

Conclusion:

By checking RAM compatibility, reseating the modules, and adjusting BIOS settings, RAM mismatch errors can be resolved and the computer can work properly.

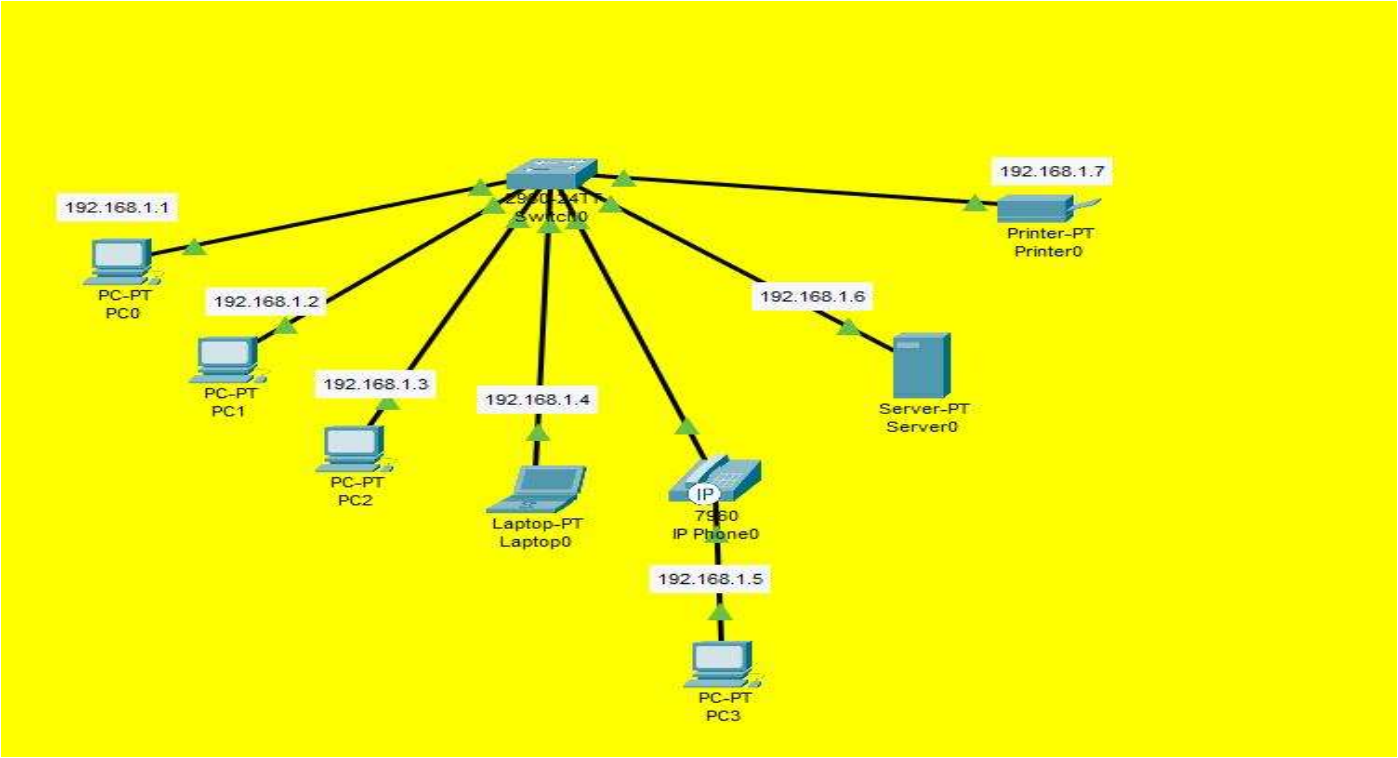
Task 3: LAN Topology Build

Build a LAN topology in Packet Tracer for 5 systems and 1 printer using a Class C subnet.

- Use Cisco Packet Tracer to build the topology.
- Include 5 end-systems and 1 network printer.
- Assign IP addresses from a Class C subnet to all devices.
- All assigned IPs must be reachable via Ping.

The following steps are used to build a LAN topology in Cisco Packet Tracer:

1. Add Devices:



- Add 5 PCs.
- Add 1 Network Printer.
- Add 1 Cisco 2960 Switch.

2. Connect Devices:

- Connect all 5 PCs and the printer to the switch.
- Use Copper Straight-Through cables.

3. Assign IP Addresses:

Use the Class C network 192.168.1.0/24.

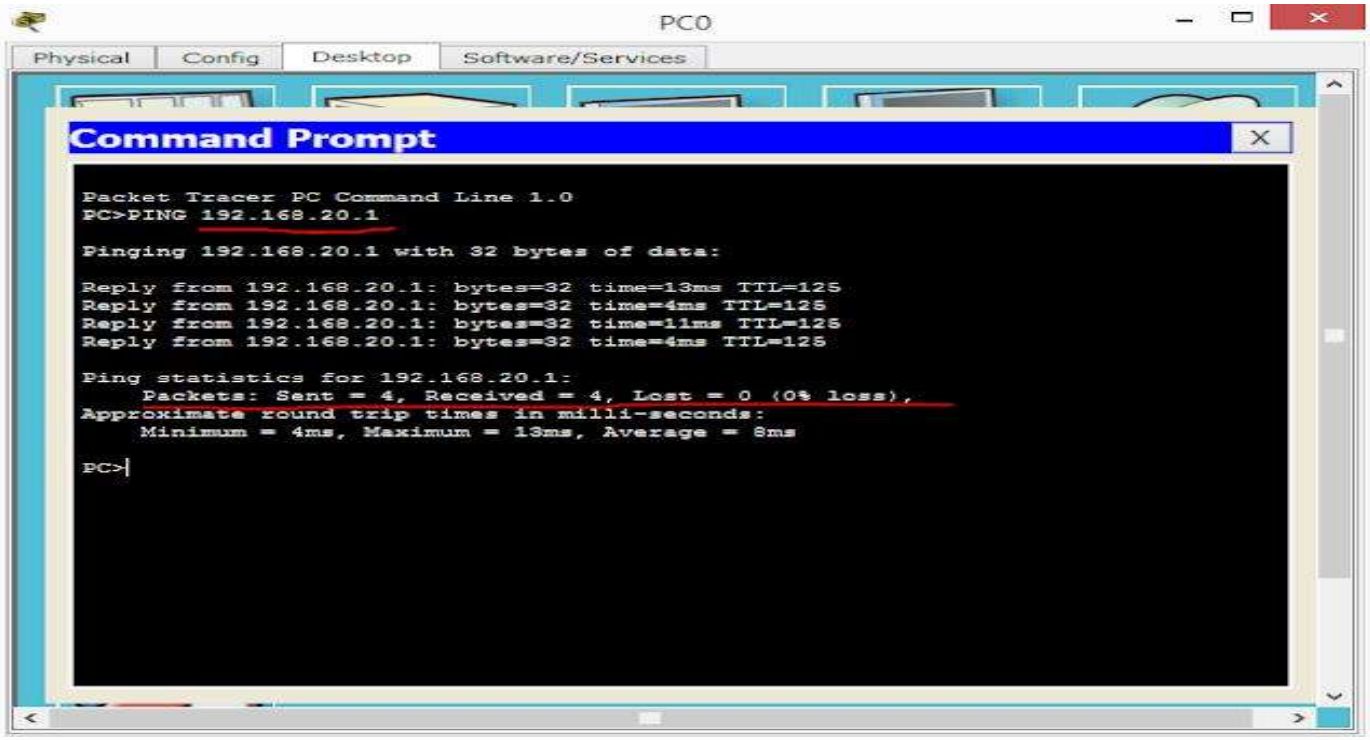
Device	IP Address
PC0	192.168.1.1
PC1	192.168.1.2
PC2	192.168.1.3
PC3	192.168.1.4
PC4	192.168.1.5
Printer	192.168.1.6

Subnet Mask for all devices: 255.255.255.0

4. Configure Devices:

- Open each PC.
- Go to Desktop → IP Configuration.
- Enter the IP address and subnet mask.
- Configure the printer with its IP address through the Config tab.

5. Perform Ping Test:



- Open Command Prompt on any PC.
- Ping the other PCs and printer.
- Example: ping 192.168.1.6
- Check for successful replies.

Conclusion:

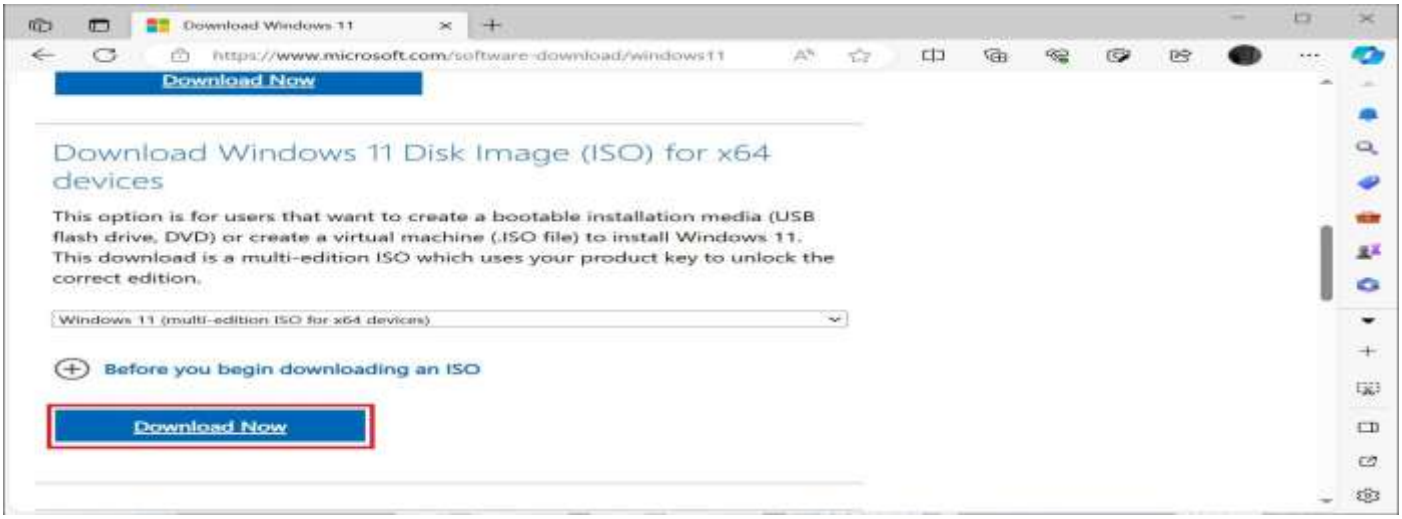
The LAN topology is successfully built with 5 PCs and 1 printer using a Class C subnet. All devices are connected through a switch and can communicate with each other through Ping.

Task 4: Clean OS Installation

Perform a clean OS installation with manual disk partitioning and startup optimization on a startup office workstation.

- Perform a clean install of Windows/Linux from ISO.
- Manually partition the disk (not guided/automatic partitioning).
- Optimize startup programs/services for faster boot. OS must boot without errors after installation.

1. Download OS ISO:



Download the Windows or Linux ISO file from its official website.

2. Create Bootable USB:



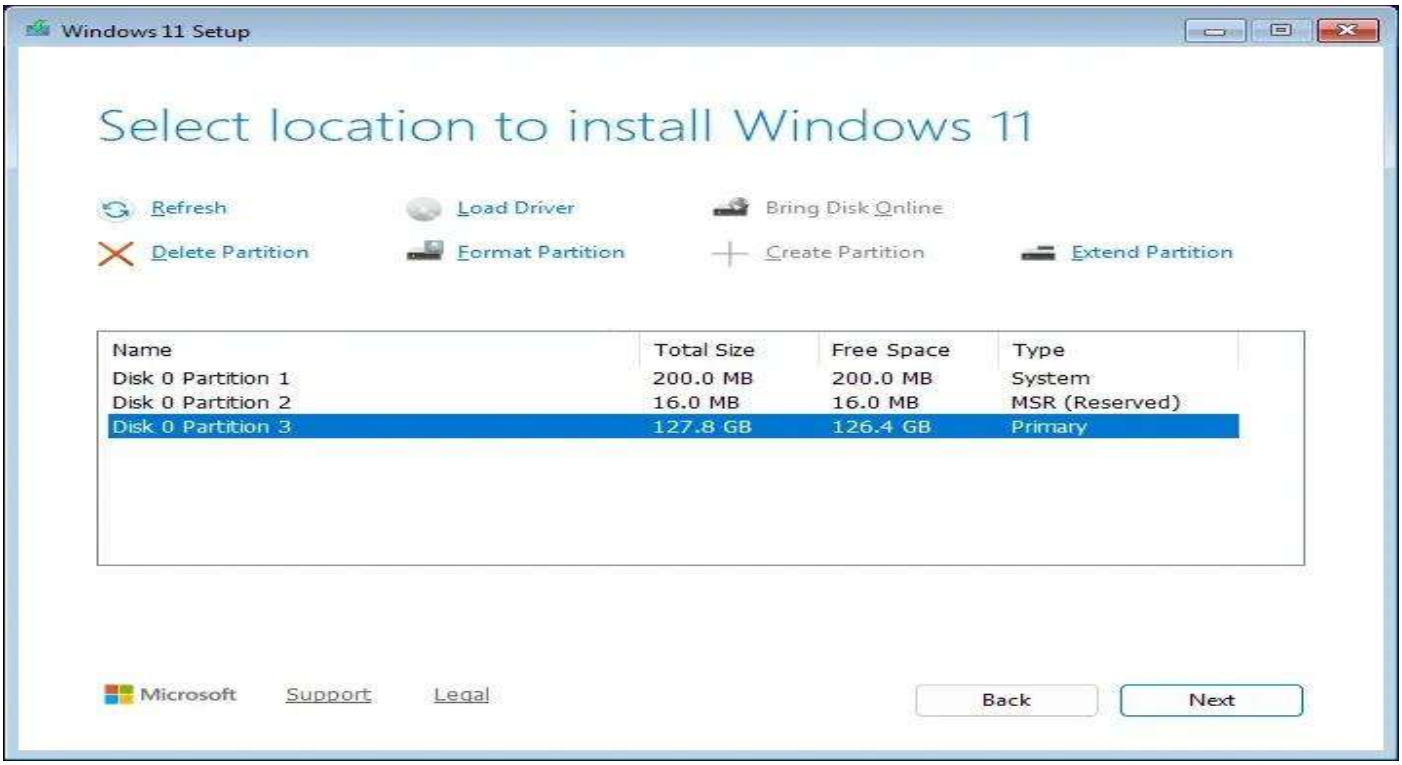
Use a tool like Rufus to create a bootable USB drive.

3. Boot from USB:



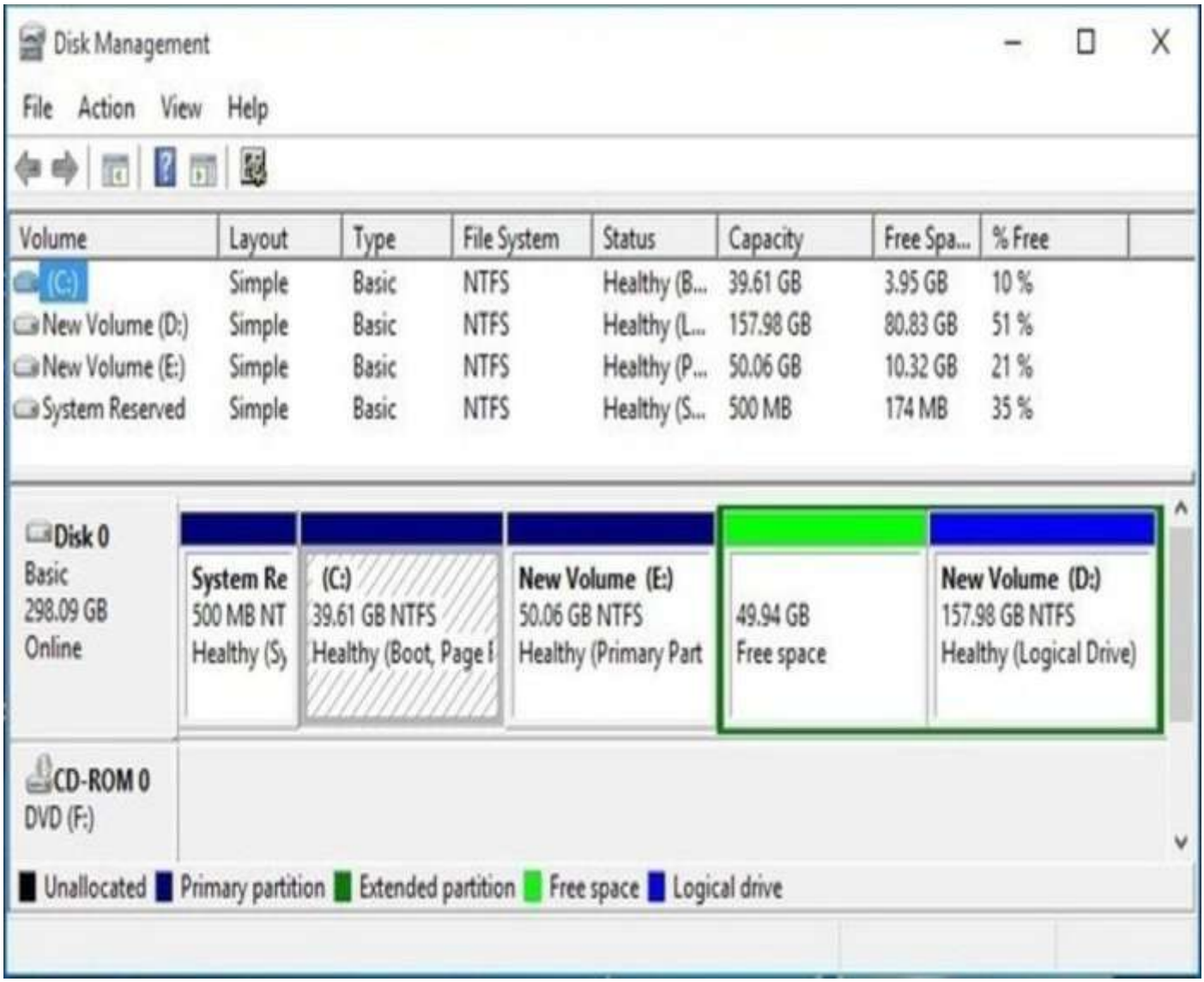
Connect the USB drive and start the computer using the USB boot option in BIOS/Boot Menu.

4. Install the OS:



Start the clean installation process and select the Custom or Manual Installation option.

5. Manual Disk Partitioning:



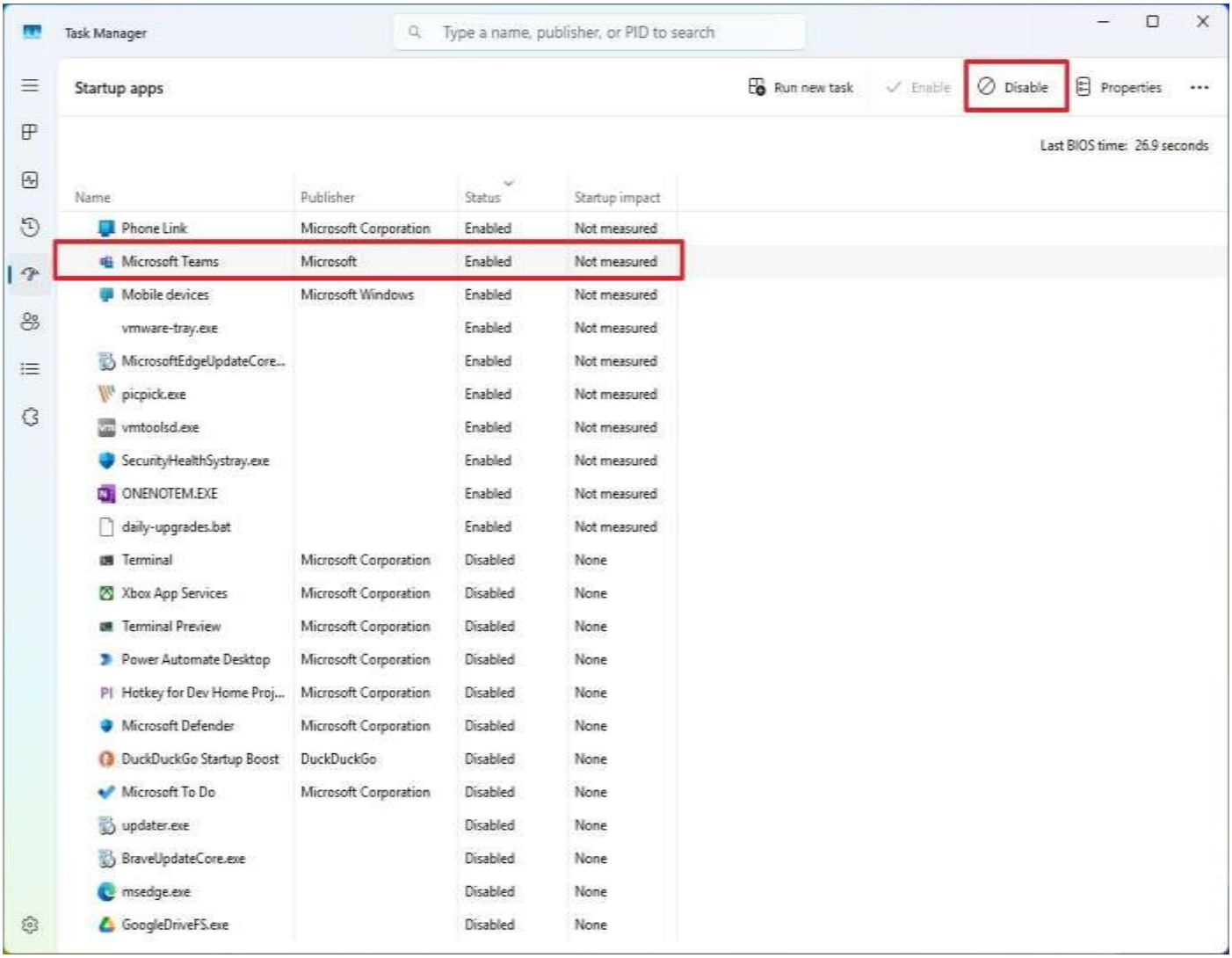
Create partitions manually according to your requirements. For example:

- C Drive: 150 GB – Operating System
- D Drive: 200 GB – Personal Files
- E Drive: 150 GB – Software and Projects

6. Install Drivers and Updates:

After installation, install the required hardware drivers and OS updates.

7. Optimize Startup:



Open Task Manager → Startup Apps and disable unnecessary startup applications. Do not disable essential system services.

8. Test the System:

Restart the computer and check that the OS boots without errors and all partitions and devices are working properly.

Conclusion:

The operating system is successfully installed with manual disk partitioning. Unnecessary startup programs are disabled for faster booting, and the system starts without errors.

Section C — Mini Capstone Project

Mini Project: Nexus Innovations — Diagnosing Random System Reboots

Objective: At Nexus Innovations, three rendering workstations pass POST every morning but restart randomly during heavy afternoon workloads. The computers are kept inside poorly ventilated desk cabinets. Analyze the probable hardware and thermal causes and propose a remediation plan, combining hardware, power, and troubleshooting concepts from this module.

Your project must:

- Identify the three hardware components most likely to cause random reboots under load.
- Explain why the systems pass POST in the morning but fail in the afternoon.
- Describe how you would use a stress-test tool to confirm a thermal or power issue.
- Propose three specific steps to remediate the frequent system restarts.
- Provide a brief technical justification for why a PSU upgrade might (or might not) be necessary.

Mini Project: Nexus Innovations — Diagnosing Random System Reboots

1. **Three Hardware Components Most Likely to Cause Random Reboots**

1. **CPU (Processor):**



During heavy rendering, the CPU produces more heat. Overheating can cause automatic shutdowns or restarts.

2. **GPU (Graphics Card):**



Rendering puts heavy load on the GPU. High temperature or unstable power can cause system instability.

3. **PSU (Power Supply Unit):**



During heavy workloads, CPU and GPU require more power. An insufficient or faulty PSU can cause random restarts.

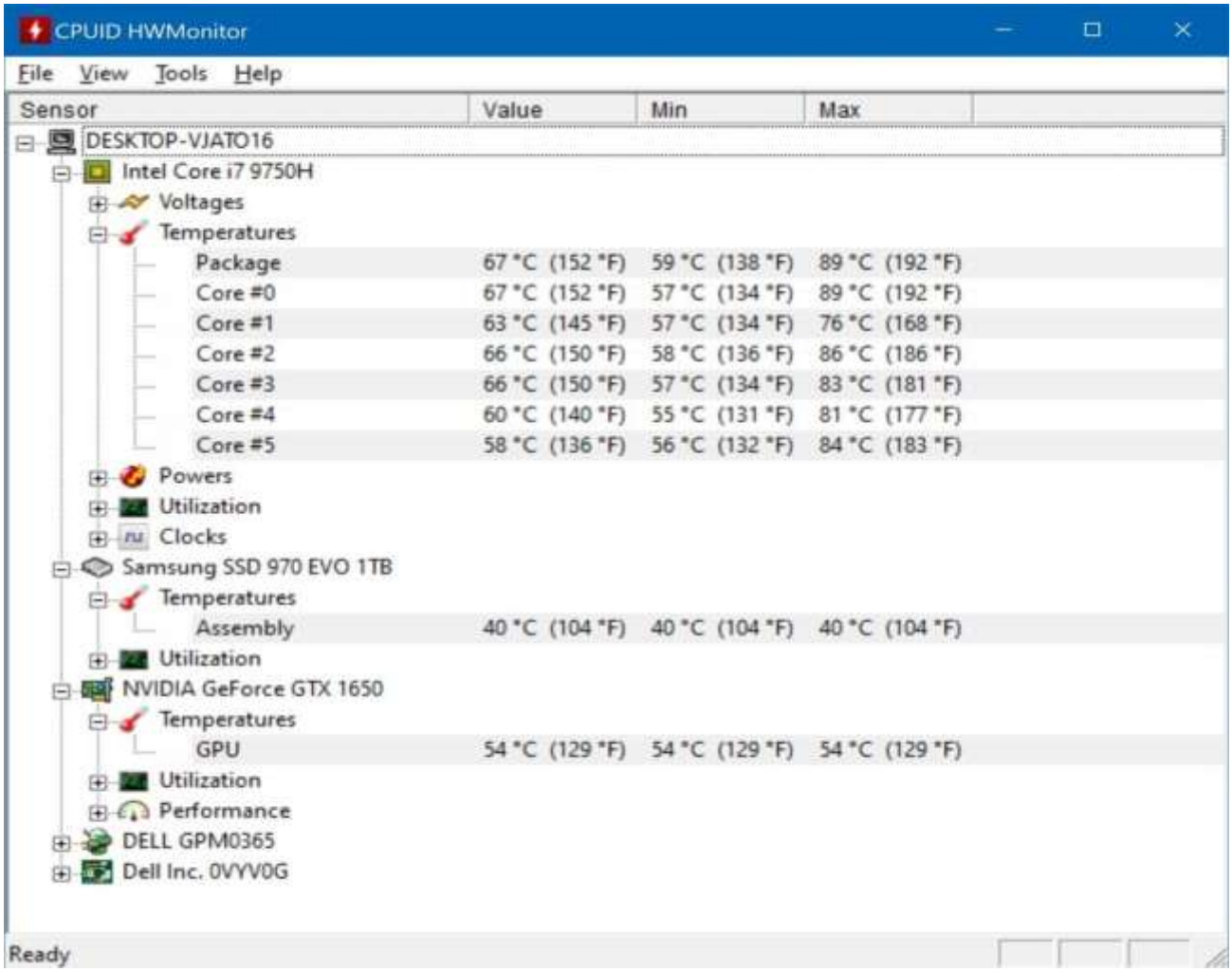
2. Why Systems Pass POST in the Morning but Fail in the Afternoon

In the morning, the computers are cool, so they pass the POST successfully. During afternoon rendering, CPU and GPU temperatures increase. Poor ventilation inside desk cabinets traps hot air and causes overheating. High power consumption can also expose PSU problems, resulting in random restarts.

3. Using a Stress-Test Tool

I would use HWMonitor and OCCT.

- Open HWMonitor to monitor CPU and GPU temperatures and voltage.



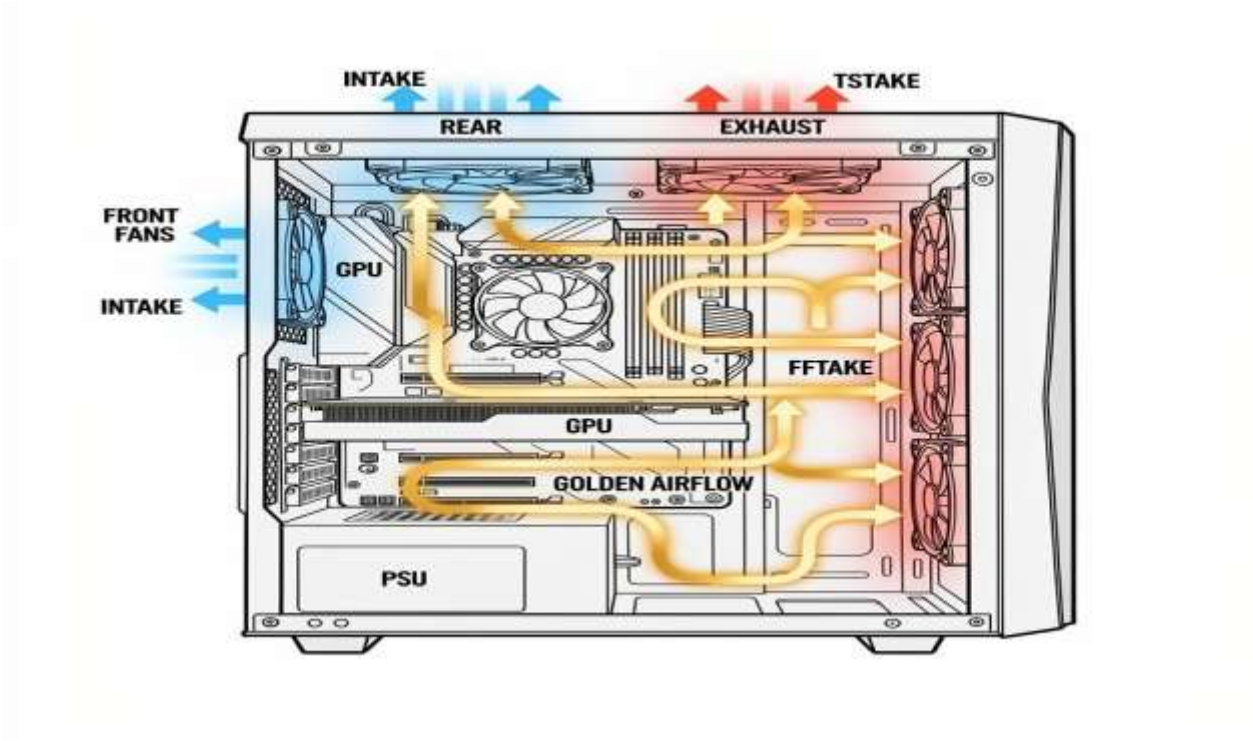
- Run CPU and GPU stress tests using OCCT.



- Observe temperature and system stability during testing.
- If the temperature becomes too high or the system restarts under load, investigate overheating or power-related issues.

4. Three Steps to Remediate Frequent System Restarts

1. **Improve Airflow:**



   Provide proper ventilation, clean dust, and install additional cabinet fans.

2. **Improve Cooling:**



Check CPU and GPU fans, clean heatsinks, and replace thermal paste if required.

3. **Check Power Supply:**



Inspect PSU capacity, cables, and connectors. Replace the PSU if it is faulty or insufficient.

5. Technical Justification for PSU Upgrade

A PSU upgrade is necessary if the existing PSU cannot provide sufficient and stable power to the CPU and GPU during heavy rendering. However, if the PSU is working properly and has sufficient capacity, an upgrade is not necessary. In that case, improving cooling and ventilation may solve the problem.

Conclusion

The most probable causes of random system restarts are CPU overheating, GPU overheating, and PSU-related power instability. Proper airflow, effective cooling, and power supply inspection can improve system stability and prevent frequent restarts.

Section D — AI-Augmented Learning

Step -1

BUILD WITH AI Use an AI

tool of your choice (ChatGPT, Claude, GitHub Copilot, etc.) to generate an IP addressing and subnetting plan for Nexus Innovations' startup office network, covering 5 systems and 1 printer.

- The AI-generated plan must specify the subnet, subnet mask, usable host range, and broadcast address.
- The plan must allocate addresses for all 5 systems and 1 printer named in the scenario.
- Instruct the AI to also identify which addresses to reserve for future expansion.
- Document the exact prompt(s) used.

BUILD WITH AI

IP Addressing and Subnetting Plan for Nexus Innovations

1. Objective

To use an AI tool (ChatGPT) to generate an IP addressing and subnetting plan for Nexus Innovations' startup office network, containing 5 computers and 1 network printer.

2. AI Tool Used

ChatGPT

3. Exact Prompt Used

"Create an IP addressing and subnetting plan for Nexus Innovations, a startup office network containing 5 computers and 1 network printer.

Use a suitable private IPv4 subnet.

Include the following details:

1. Network subnet (CIDR notation).
2. Subnet mask.
3. Total and usable IP addresses.
4. Network address.
5. Usable host range.
6. Broadcast address.
7. Assign a unique IP address to all 5 systems and 1 printer.
8. Identify the IP addresses reserved for future expansion.
9. Reserve one IP address for the default gateway.

Present the plan in clear tables with a simple technical explanation."

4. Network Subnet Details

Parameter	Value
Network Subnet	192.168.1.0/28
Subnet Mask	255.255.255.240
Total IP Addresses	16
Usable Host Addresses	14
Network Address	192.168.1.0
Usable Host Range	192.168.1.1 – 192.168.1.14
Broadcast Address	192.168.1.15

5. IP Address Allocation

Device Name	Assigned IP Address
System 1	192.168.1.2
System 2	192.168.1.3
System 3	192.168.1.4
System 4	192.168.1.5
System 5	192.168.1.6
Network Printer	192.168.1.7

Default Gateway: 192.168.1.1

6. Future Expansion

The following IP addresses are reserved for future devices:

IP Address	Purpose
192.168.1.8 – 192.168.1.14	Future computers, servers, and network devices

7. Technical Explanation

The network uses a /28 subnet, which provides 16 total IP addresses and 14 usable host addresses.

One IP address is reserved for the default gateway, six IP addresses are assigned to five computers and one printer, and seven usable IP addresses remain available for future expansion.

The network address and broadcast address cannot be assigned to normal devices.

8. Conclusion

Using ChatGPT, an IP addressing and subnetting plan was successfully generated for Nexus Innovations. The 192.168.1.0/28 subnet provides sufficient IP addresses for all five computers and one printer, with additional addresses reserved for future network expansion.

Step-2

TEST & DEBUG (WITHOUT AI)

Then, working without AI, verify the plan for errors — overlapping ranges, incorrect network/broadcast addresses, insufficient host capacity, or unreserved growth space — and correct them yourself.

SUBMIT

1. The exact prompt(s) you gave the AI tool.
2. The AI's original subnetting plan and your corrected version.
3. A 3–4 line note explaining what you changed and why the AI's version needed it.

TEST & DEBUG (WITHOUT AI)

IP Addressing Plan Verification – Nexus Innovations

1. Exact Prompt Given to AI

"Create an IP addressing and subnetting plan for Nexus Innovations, a startup office network containing 5 computers and 1 network printer.

Use a suitable private IPv4 subnet. Include the subnet, subnet mask, usable host range, network address, broadcast address, IP allocation for all devices, default gateway, and addresses reserved for future expansion. Present the plan in a clear table."

2. AI's Original Plan

Original Subnet Details:

Parameter	Value
Network	192.168.1.0/28
Subnet Mask	255.255.255.240
Network Address	192.168.1.0
Usable Host Range	192.168.1.1 – 192.168.1.14
Broadcast Address	192.168.1.15
Usable Hosts	14

Original IP Allocation:

Device	IP Address
Router	192.168.1.1
System 1	192.168.1.2
System 2	192.168.1.3
System 3	192.168.1.4
System 4	192.168.1.5
System 5	192.168.1.6
Printer	192.168.1.7
Future Devices	192.168.1.8 – 192.168.1.14

3. Corrected Version

After manually checking the subnet and IP allocation, I verified that the original plan was technically correct.

The corrected plan keeps the same subnet and IP addresses but clearly separates the assigned and reserved addresses.

Device	IP Address	Purpose
Router	192.168.1.1	Default Gateway
System 1	192.168.1.2	Computer
System 2	192.168.1.3	Computer
System 3	192.168.1.4	Computer
System 4	192.168.1.5	Computer
System 5	192.168.1.6	Computer
Printer	192.168.1.7	Network Printer
Future Devices	192.168.1.8 – 192.168.1.14	Reserved

Final Subnet Details:

- Subnet: 192.168.1.0/28
- Subnet Mask: 255.255.255.240
- Network Address: 192.168.1.0
- Usable Host Range: 192.168.1.1 – 192.168.1.14
- Broadcast Address: 192.168.1.15
- Total Usable Hosts: 14

4. Manual Verification Note (3–4 Lines)

I manually checked the AI-generated subnetting plan without using AI. I verified that there were no duplicate IP addresses and the network and broadcast addresses were correct. The subnet provides 14 usable addresses, which are sufficient for all devices. I clearly reserved the remaining 7 addresses for future expansion, so no major correction was required.

5. Conclusion

The IP addressing plan was successfully tested and verified manually. The subnet has sufficient host capacity, no overlapping IP addresses, correct network and broadcast addresses, and enough reserved addresses for future expansion.